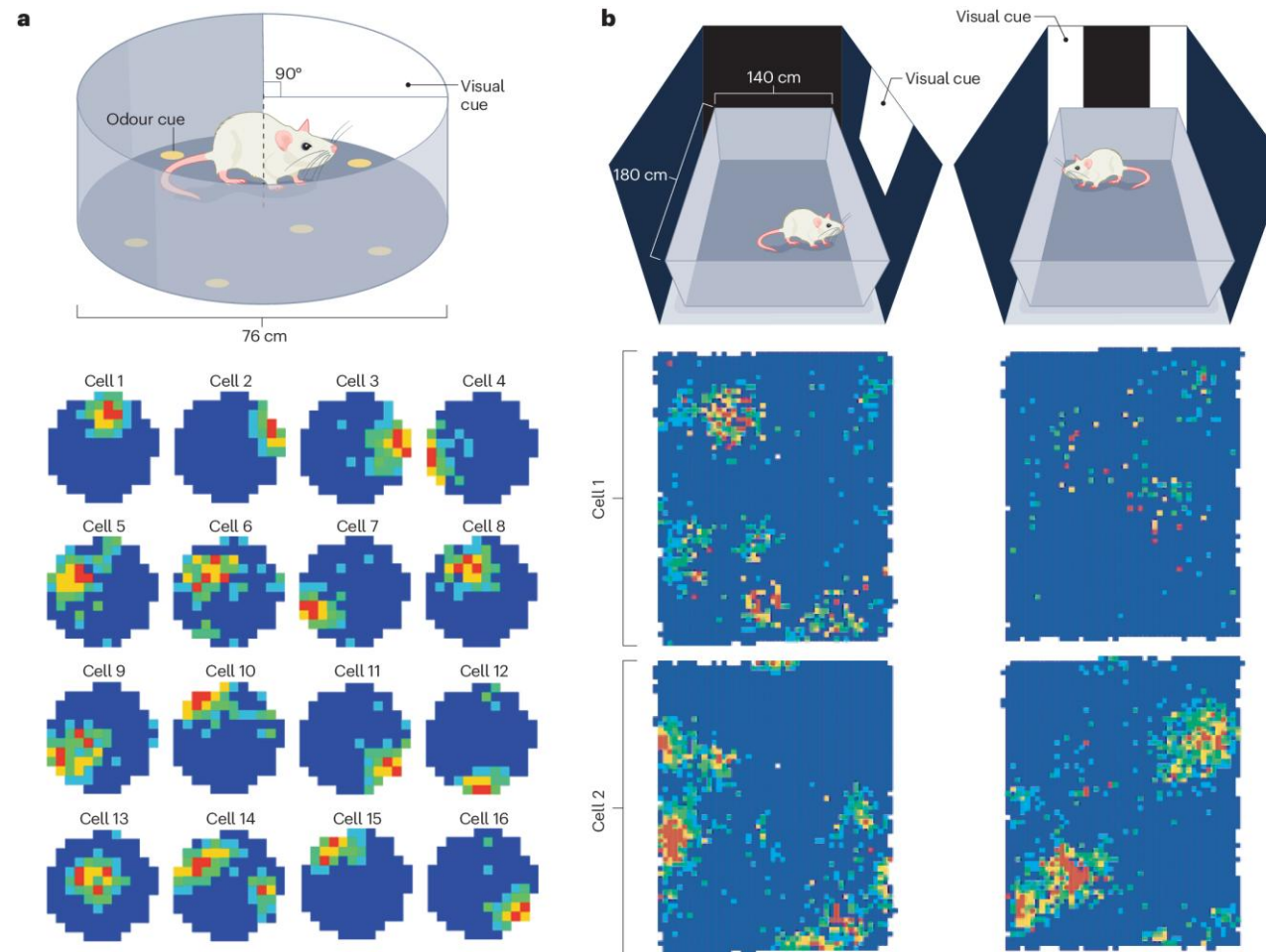


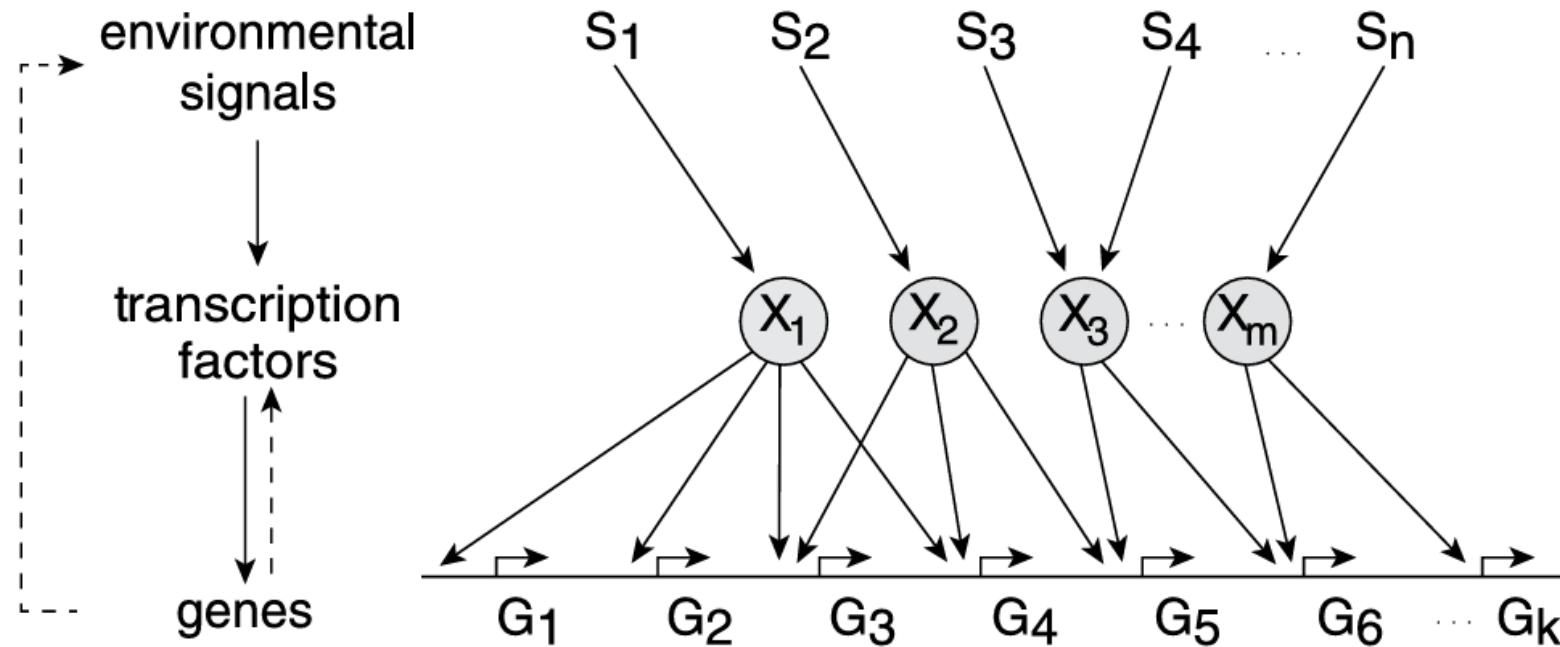
Biological systems represent external environments in internal state spaces



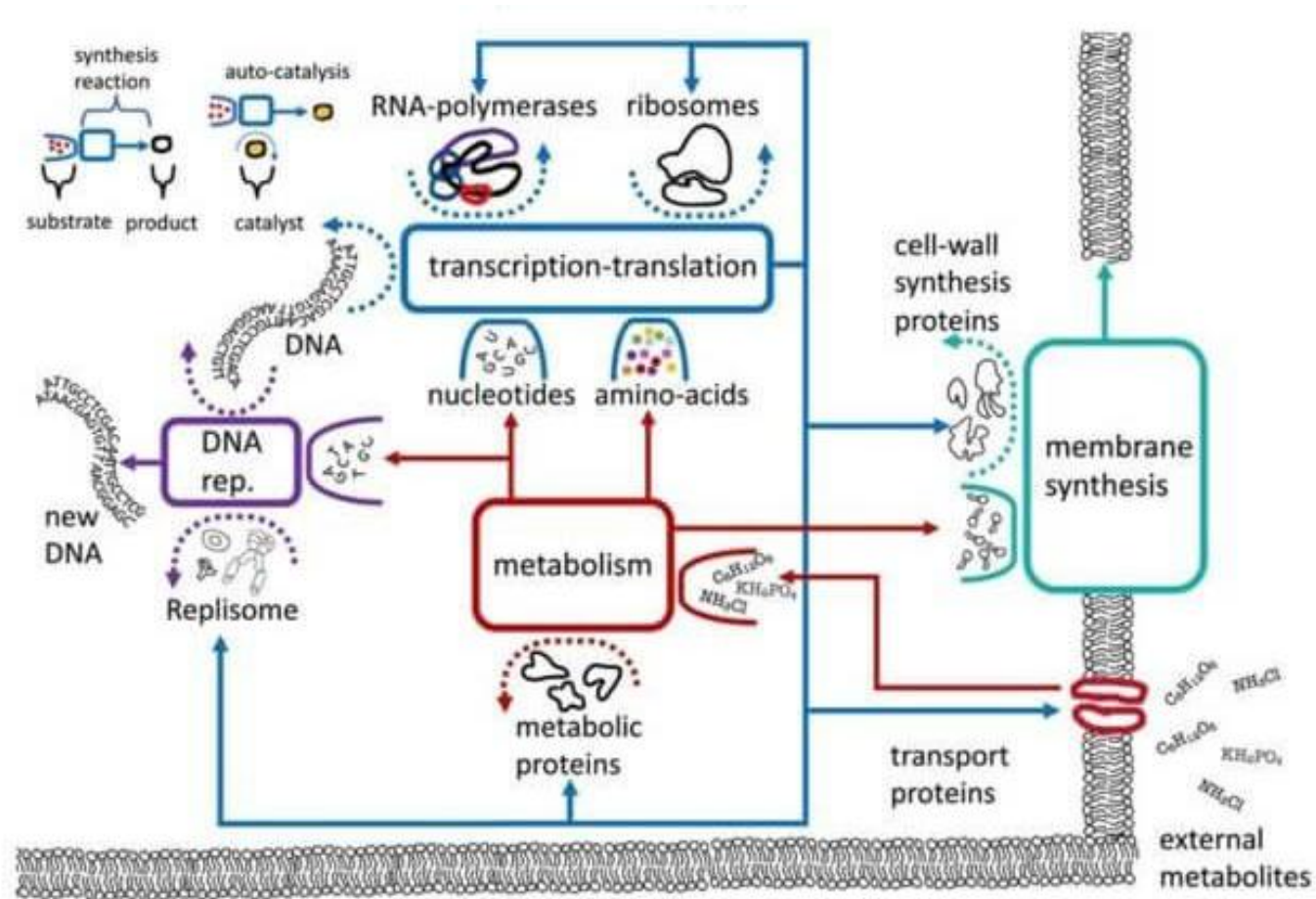
“Every good regulator of a system must be a model of that system.”

Conant & Ashby (1970)

E. coli represents the external environment inside the cell via about 300 special proteins called transcription factors (TFs)

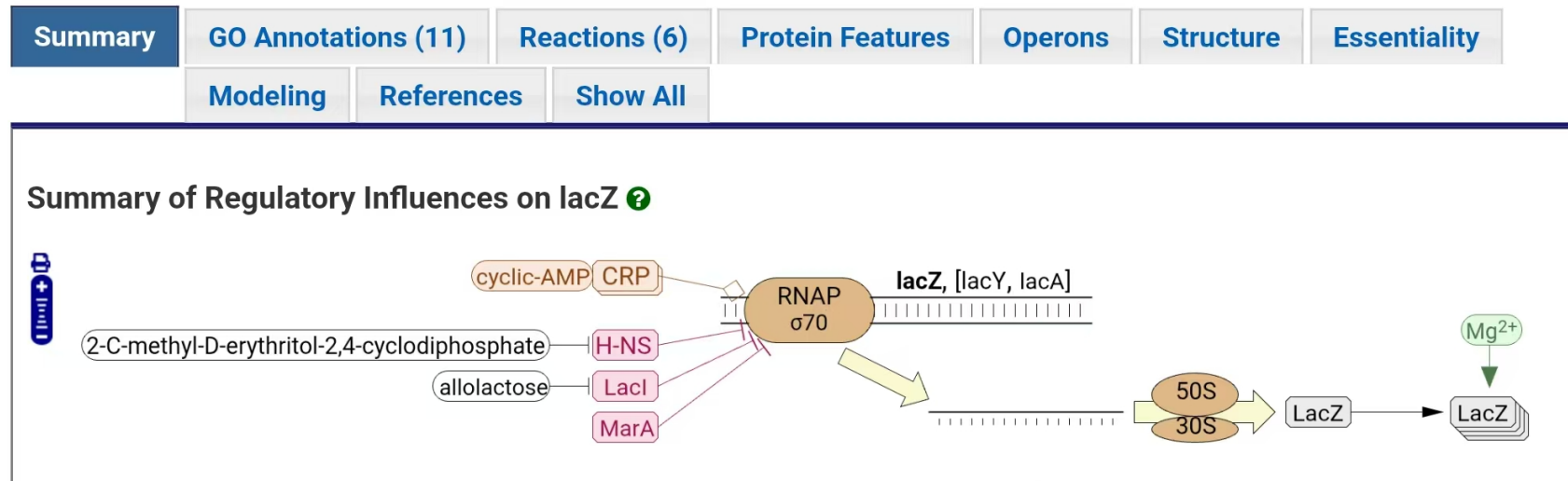


TFs coordinate the chemical brain of a self-reproducing machine

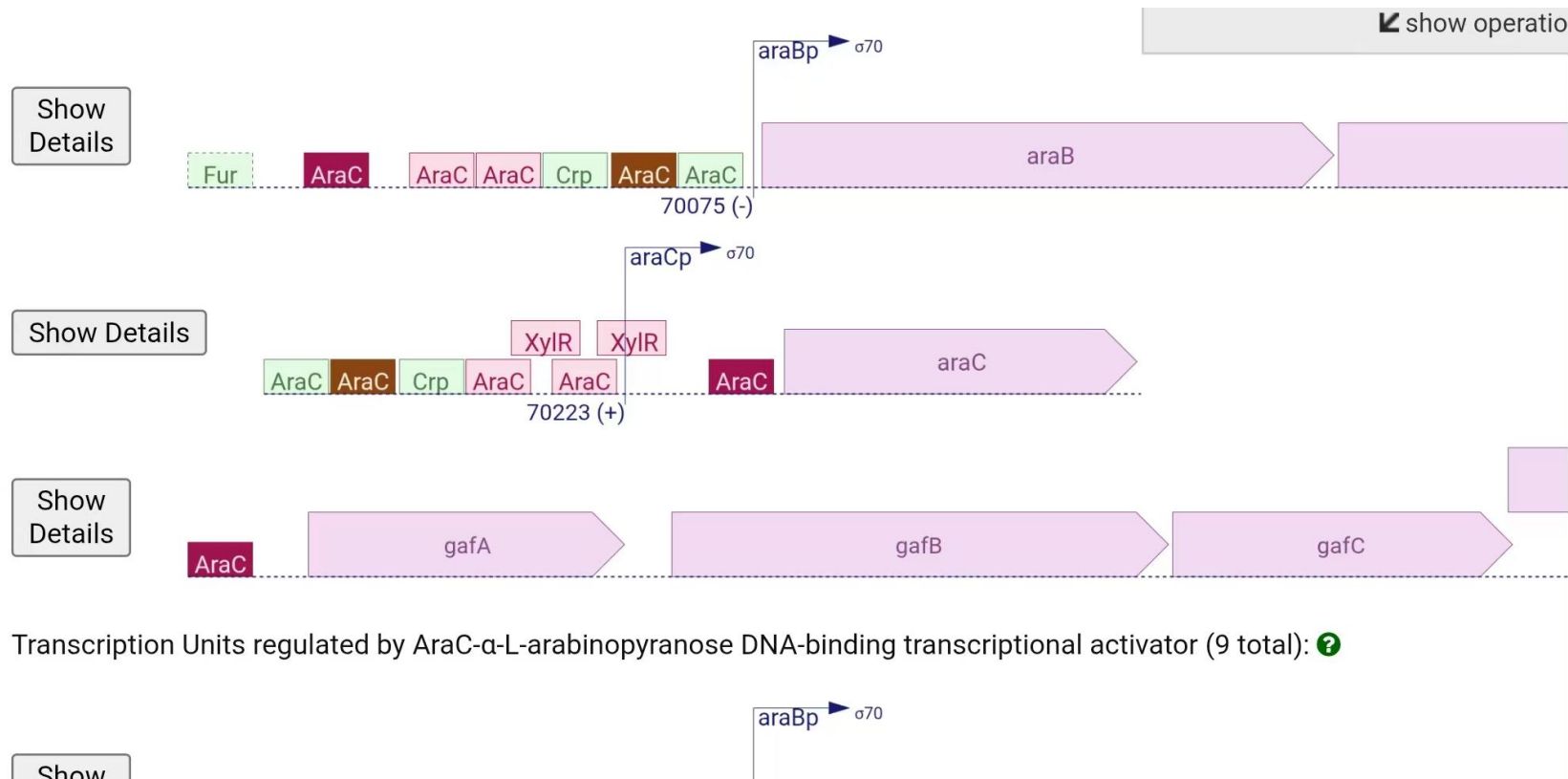


Source: Anjan Roy et al., PNAS 2021; 118(33):e2107829118

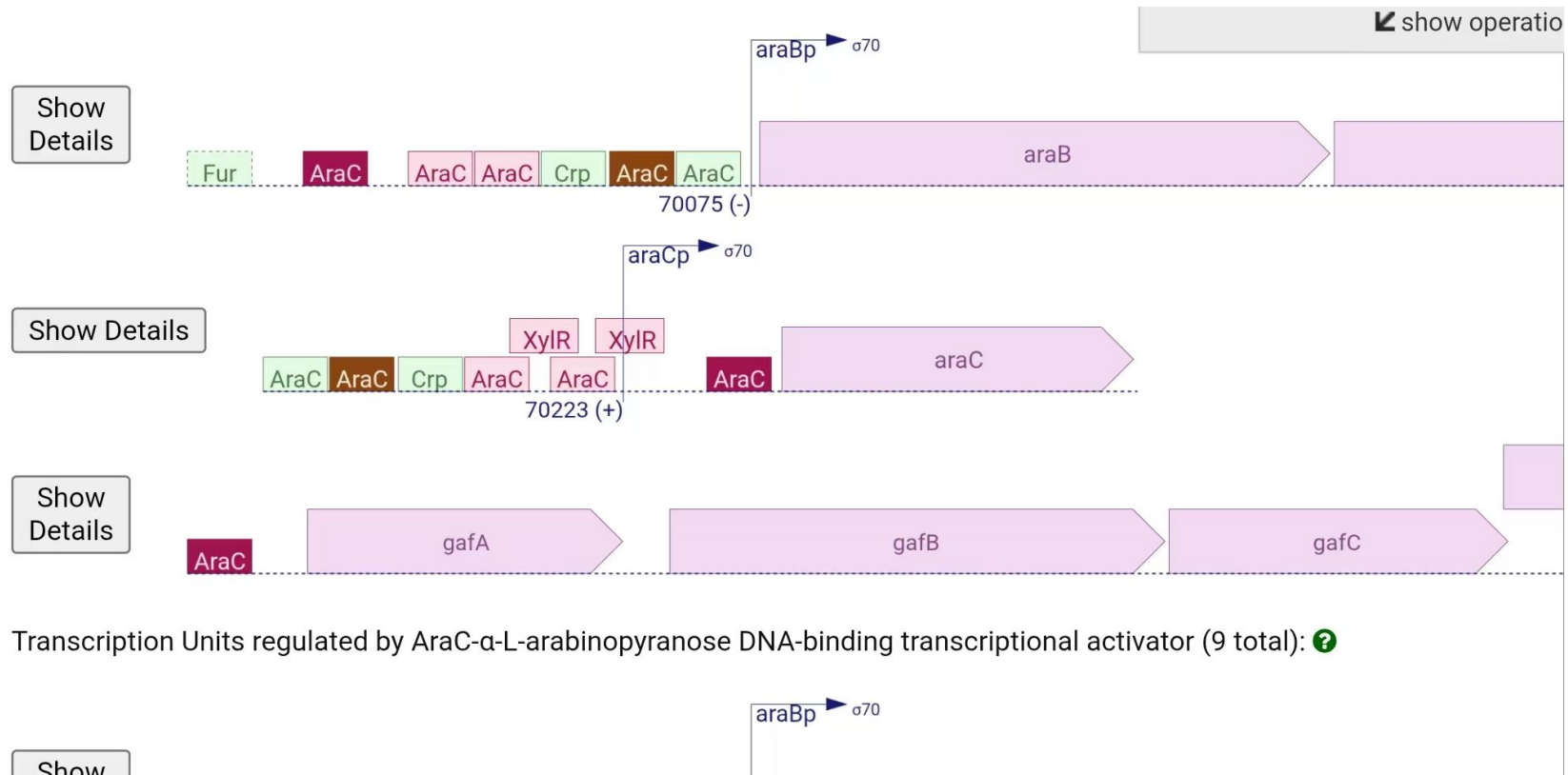
The *E. coli* regulatory network was reconstructed from mechanistically defined interactions



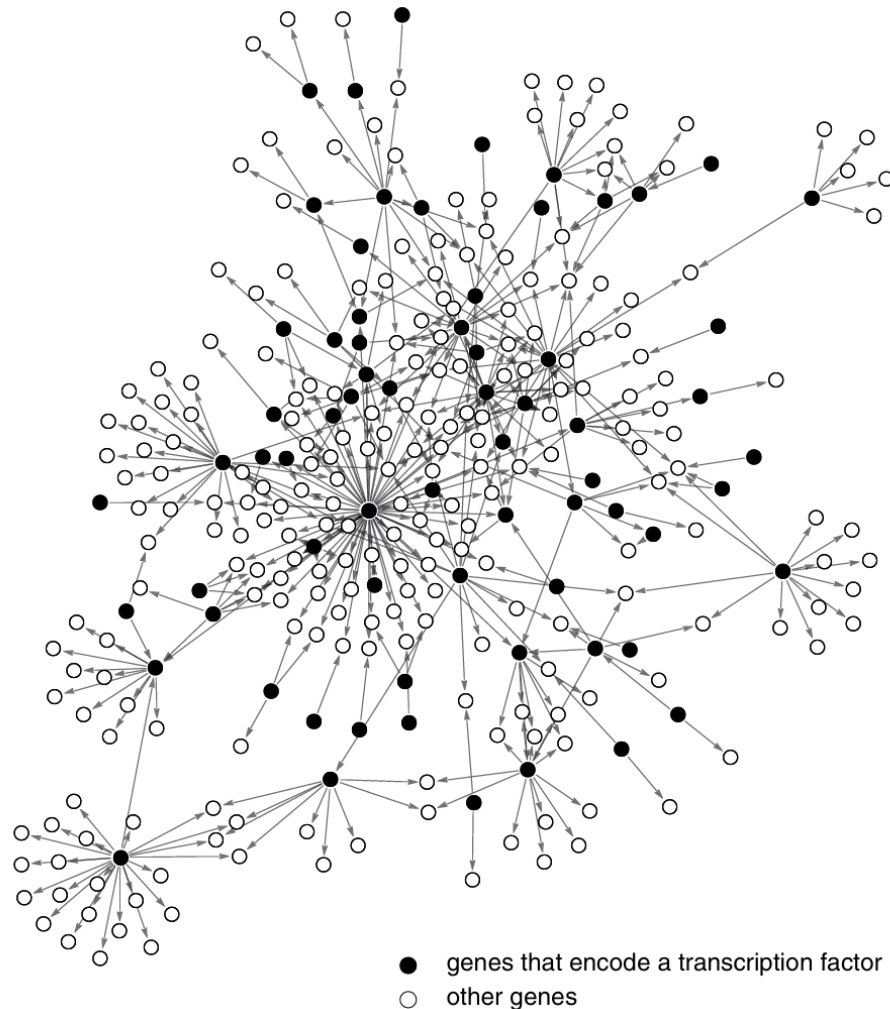
The *E. coli* regulatory network was reconstructed from mechanistically defined interactions



Notice the fact that araC control both functional genes and itself and is controlled by other TFs



Gene regulatory networks (GRNs) provide a genome-wide view of cellular cognition



- $N=4,500$ nodes-genes
- $E = 10,000$ arrows

We make the following simplifications in the rest of Block1, to understand the structure-functions of GRNs

- Track cellular internal states by a small number of protein concentrations (TFs)
- Preserve the topology of GRN, each arrow is a little miracle

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- Focus on timescales slower than 1hour
- Convert the fast biochemical reactions into effectively instantaneous input-output functions

How do we model the dynamics of a gene regulatory network?

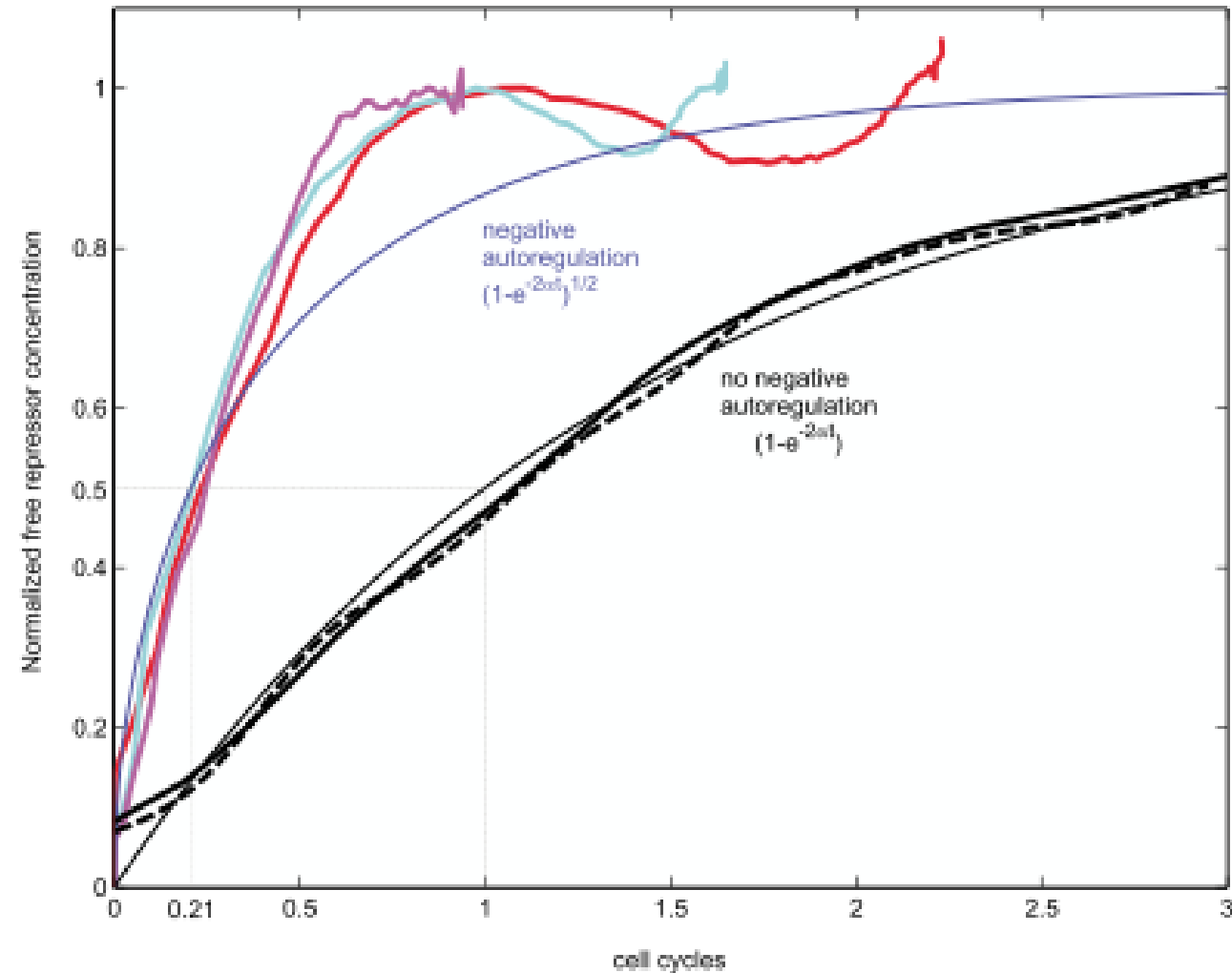
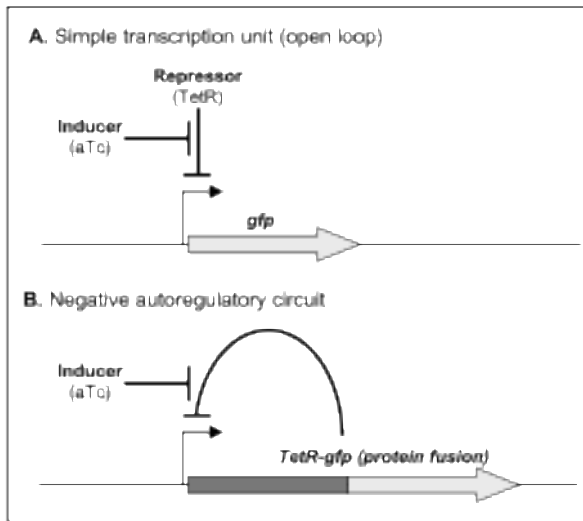
TABLE 1.2 Typical Timescales for the Reactions in the Transcription Network of *E. coli*

Binding of a small molecule signal to a transcription factor, causing a change in its activity	~1 msec
Binding of an active transcription factor to its DNA site	~1 sec
Transcription and translation of the gene	~minutes
Timescale for 50% change in concentration of the translated protein (stable proteins)	~1 hour (1 cell generation)

We make the following simplifications in the rest of Block1, to understand the structure-functions of GRNs

- Preserve the topology of GRN, each arrow is a little miracle
- Track cellular internal states by a small number of protein concentrations (TFs)
- Focus on timescales slower than 1hour
- Convert the fast biochemical reactions into effectively instantaneous input-output functions
- Work initially under balanced-growth conditions

Negative Autoregulation accelerate response time



Negative Autoregulation reduce cell-to-cell variations

